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Paper 4: Selection Functionals Reviews Response

Ezequiel Boero <ezequiel.boero@adagioinstitute.org> Wed, Sep 23, 2026 at 6:08 PM
To: "Benjamin D. Summers" <bsummers@adagiollc.com>, "ISMAIL-SUTTON, SARA R." <sara.r.ismail-sutton@durham.ac.uk>

Ben, hi again;

My assessment of this paper is favourable in substance, subject to two focused amendments set out at the end.

Your correction of my Finding 2 is right. My equation (1) presented $1/\sigma_eff^2 = 1/\sigma_\Pi^2 + \lambda/\sigma_v^2$ as exact, and it is not.
From the interior-row matching, $v - 2u = (1-a_\Pi)^2/\sigma_\Pi^2 + \lambda(1-a_v)^2/\sigma_v^2$, which is exactly $\delta t^2(k_\Pi^2/\sigma_\Pi^2 + \lambda k_v^2/\sigma_v^2)$, so the exact relation is the one in k^2/σ^2
and mine is its leading order.
On Example I the exact relation holds with residual $2.4\cdot10^{-13}$ while mine is off by 0.0033, and on refining δt from 0.02 to 0.0002 the ratio of that discrepancy to
δt sits at 0.165, 0.163, 0.162, 0.162 (first order, not rounding). Corollary 3.6(ii) states both with the $O(\delta t)$ where it belongs.
My report should have done the same.

Verification of the new material. I re-derived it from the definitions rather than checking your arithmetic against mine.

- **Corollary 3.6 on Example I.** $\sigma_eff^2 = 0.499177$, $k_eff = 1.5798$, and mean interior increment variance 0.01010 under
  Q against 0.02012 under Π. $Var_\Pi(\xi_M) = 0.438071$ and $\chi^2(Q\|\Pi) = 1372.4$, both exact.
  The asymptotic variances reproduce: 30.8925 at k = 2, 4.9987 at k = 12.39, 20.3342 at k = 2.76.
- **Calibration.** The closed form gives $k^* = 2.7618$ with the sum over $m = 0,\ldots,M-1$, and the kinetic sum alone minimises at 1.000.
- **Lemma 8.1.** $\lambda c_2 \delta t\|\Sigma_\Pi\| = 0.4519$ at $\lambda = 3/4$ and 0.6025 at $\lambda = 1$, with $\|\Sigma_\Pi\| = 10.04$.
- **Example 7.4.** Least eigenvalues 0.3387 and 1.2703; at the triple (3,5,7) the direct coefficient is −0.008588 against a composed 0.305127,
  both exact, with gap 0.314; none of the 56 triples composes and the smallest gap is 0.011, at (6,7,8). And $K_\Pi = K_target - \lambda K_v$ identically.
  The sentence you added (that zero bias is attained by presupposing the target law) is the right delimitation of what my construction shows,
  and neither review made it.
- **Example II.** I rebuilt the double-well benchmark by transfer-matrix quadrature on a mesh of my own.
  With the divergence term $F^* = -0.547760$, $\chi^2 = 162.1$, $\sigma^2 = 44.31$; without it, −0.637163, 169.3, 26.60; the proposal's value −0.561929;
  the term shifts the estimand by 0.089 and reverses the sign of the displacement. All six figures reproduce to every printed digit.
  That answers Finding 6 completely, and the sign reversal is a stronger result than the finding asked for.

One caution for anyone reading the deposit as extracted text rather than as a typeset page: the display of K_Π in Example 7.4 comes out of the
PDF with ε inside the $1/\delta t$ factor, and under that reading the proposal has least eigenvalue −15.42 and is not a probability measure.
The typeset page has ε outside and everything checks. I mention it because I nearly reported it as an error.

The two Monte Carlo figures. You are right that I cannot be relied on here, and I ran the study a third time with a seed belonging to neither of us:
1000 replications at $N = 10^4$ give a pooled $\sqrt{N}\cdot$std of 3.817 with bootstrap standard error 0.127, and a range of 2.96 to 4.99
over twenty disjoint batches of fifty.

On the 7.53: my maximum over four thousand draws of fifty-replication batches is 5.79, and yours over two hundred batches is 6.88.
My figure sits above both, and none of my four thousand simulated batches reached it.
I accept it as an artefact of my implementation.

On the 3.44: I would not settle this in either direction. The three estimates order as 3.44 < 3.82 < 4.02 with standard errors too small for the
gaps to be sampling noise, which points to unresolved implementation differences and does not, by itself, identify any one implementation as the source.
What the episode shows is what Finding 7 was about: the statistic is not reproducible across implementations without deposited code,
and you have deposited script, seeds and output. The substance survives in all three runs (my pooled value is 31% below the asymptote 5.558,
you report 28%, and all three of us find the spread across fifty-replication batches larger than the effect the July caption was interpreting).

On the remedy you did not adopt. I accept the refusal and the reason. Subtracting a reference kinetic term does convert the weight into a
Girsanov ratio and the paper into an account of an established method, and preserving the construction while stating exactly
what it is was the better choice.

Two amendments, both narrow.

1. Proposition 7.5 needs a uniform argument, and the Gaussian route supplies one. The proof says the backward Riccati recursion converges
geometrically to the interior fixed point, so all but a bounded number of increments have $v_m = \sigma_{eff}^2(1+o(1))$.
Because $\delta t = T/M$ tends to zero, geometric convergence at each fixed δt does not give a bounded layer: the contraction factor approaches one.
I measured it on the Example I pair. The local contraction factor at the fixed point is 0.9378, 0.9961, 0.9990 and 0.9995 at M = 50, 800, 3200 and 6400,
and the number of steps for the relative deviation to fall below 10^{-6} exceeds the whole grid for every M up to 3200, reaching 5189 steps at M = 6400,
which is 0.81M.
The transient occupies a fixed fraction of the horizon, so the step does not hold as written.
I would also note that a recursion run from the last index backward yields the conditional variances given the future, not the innovation variances of the forward representation the proof attributes to it.

The conclusion appears recoverable by a direct Gaussian quadratic-form argument, which needs no statement about a boundary layer at all.
Under Q the increment vector Y is centred Gaussian with covariance C_M, and the realized quadratic variation $V_M = Y^TY$ satisfies
$E_Q[V_M] = \mathrm{tr}(C_M)$ and $\mathrm{Var}_Q(V_M) = 2\mathrm{tr}(C_M^2)$, so convergence in Q-probability follows by Chebyshev from $\mathrm{tr}(C_M) \to \sigma_{eff}^2 T$ together
with $\mathrm{tr}(C_M^2) \to 0$. Both behave as required in Example I: $\mathrm{tr}(C_M)$ runs 0.504805, 0.502397, 0.501197, 0.500598, 0.500299, 0.500150
at M = 50 to 1600 against the limit 1/2, an $O(\delta t)$ error, and $2\mathrm{tr}(C_M^2)$ runs 0.010303, 0.005075, 0.002519, 0.001255, 0.000626, 0.000313,
which is $\delta t/2$ to within 3% at M = 50 and 0.1% by M = 1600.
What the proposition needs is therefore proofs of those two trace estimates for the Gaussian pairs of Corollary 3.6 (statements about a Toeplitz precision with one modified row) rather than a claim about how fast a transient dies.
I should add that the marginal increment variances, which is what I checked first, establish convergence of the mean only and not the concentration;
the second trace is what carries it.

Separately, the parenthetical that a random limiting quadratic variation gives $d_{TV} \to 1$ for every triple is too broad: if the random limit puts positive mass at σ_{eff}^2, the separation of events fails on that event.
It needs a hypothesis excluding that, or it should be dropped.

2. The abstract states the $O(\delta t)$ relation as an identity. The abstract says the selected measure acquires the diffusion coefficient $1/\sigma_{eff}^2 = 1/\sigma_\Pi^2 + \lambda/\sigma_v^2$,
while Corollary 3.6(ii) gives that relation with an $O(\delta t)$ term and reserves exactness for the relation in k^2/σ^2.
The equality without the remainder is the refinement-limit statement, not the finite-grid one.
I raise it with some awkwardness, since it is the same imprecision you corrected in my own report, but the abstract is where the formula
will be read and quoted, and it should either carry the remainder or be stated as the limit.

Neither amendment touches the construction. One completes an asymptotic argument, for which a viable route is available;
the other aligns a headline formula with the corollary that states it precisely.

Two things I note, with nothing asked. Your correction of the July Mori–Zwanzig description (that dropping the fluctuating force leaves $\sigma = 0$,
which is not a surrogate under your own Definition 3.3) was not prompted by either review, and it is the second unsupported statement you have removed
in this cycle without anyone objecting to it.

The attribution of *Stochastic optimal control matching* to Domingo-Enrich, Han, Amos, Bruna and Chen is correct. And Remark 7.6 shows how the
sequencing criterion can be applied without leaving anything hanging: you kept the one dependency that survives it, the integrated deficit from
the Koopman paper, cited, and dropped the mentions of the liftability profile and the memory rank that belonged to the removed citations.
Nothing in this manuscript is now posed as open that another member of the portfolio settles.
That is precisely the adjustment I have asked for in the memory-rank paper, where the same criterion was applied and Open Problem (v) was
left standing, I record it here only for consistency across the portfolio, and no further amendment to the present manuscript is requested on that account.

Subject to the two amendments, I regard the questions in my report as answered or correctly identified as open, with the answer to Question 16
appropriately recorded as a limitation: absent an externally established truth link, the decision-theoretic value of F*(v) remains application-specific.

Best,
Ezequiel

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